AIR UNIVERSITY

DEPARTMENT OF ELECTRICAL AND COMPUTER ENGINEERING

EXPERIMENT NO 8

Lab Title Convolution

Student Name: Ali Abbas Khan

Reg. No: 230661

Objective:

LAB ASSESSMENT:

Attributes	Excellent (5)	Good (4)	Average (3)	Satisfactory (2)	Unsatisfactory (1)
Ability to Conduct Experiment					
Ability to assimilate the results					
Effective use of lab equipment and follows the lab safety rules					

Total Marks: _____ Obtained Marks: _____

LAB REPORT ASSESSMENT:

Attributes	Excellent (5)	Good (4)	Average (3)	Satisfactory (2)	Unsatisfactory (1)
Data presentation					
Experimental results					
Conclusion					

Total Marks: _____ Obtained Marks: _____

Date: _____ Signature: _____



Department of Electrical and Computer Engineering

Signal and Systems Lab

Lab Report: 8

Title: Derivation &

Integration

Name: Ali Abbas Khan

Roll Number: 230661

Class: BCE-5B

Submitted To: Engr. Maryam

Date**:** November 19, 2025

Lab Task :

1. **Integrate the following expression:**

$$\text{Cos}\,(a * y)$$

Evaluate the numeric results for fixed variables after integration

Code:

Editor - C:\Users\Admin\OneDrive\Desktop\SNS Lab\Matlab Files\L8\T1.m

T1.m

```
1 -    syms a y
2 -    f = cos(a * y);
3 -    F_y = int(f, y)
4 -    a_sub = 2;
5 -    y_sub = sym(pi)/4;
6 -    F_exact_result = subs(F_y, [a y], [a_sub y_sub])
```

Output:

Command Window

```
>> T1

F_y =

sin(a*y)/a


F_exact_result =

1/2
```

2. **Differentiate the following expression:**

$$f(y) = e^{-2by}\, y^{4a}\, \cos(cy)$$

Simplify your answer

Code:

Editor - C:\Users\Admin\OneDrive\Desktop\SNS Lab\Matlab Files\L8\T2.m

T1.m | T2.m

```
1 -    syms y a b c
2 -    f = exp(-2*b*y) * y^(4*a) * cos(c*y);
3 -    df_dy = diff(f, y);
4 -    df_dy_simplified = simplify(df_dy);
5 -    pretty(df_dy_simplified);
6
```

Output:

Command Window

```
>> T2
   4 a - 1
- y        exp(-2 b y) (2 b y cos(c y) - 4 a cos(c y) + c y sin(c y))
```

3. **Integrate the given expression:**

$$\int_{-\pi}^{\pi} e^{-2by}\cos(cy)\, dy$$

Substitute b = 3 and c = 5 after integration to get the results.

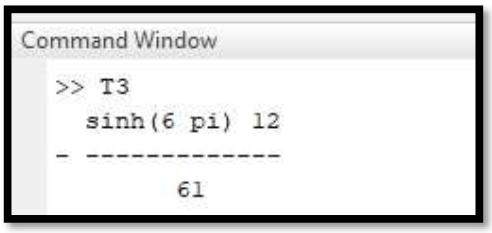
Code:

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T1.m | T2.m | T3.m

```
1 -    syms y b c
2 -    f = exp(-2*b*y) * cos(c*y);
3 -    I_definite = int(f, y, -sym(pi), sym(pi));
4 -    b_sub = 3;
5 -    c_sub = 5;
6 -    I_exact_result = subs(I_definite, [b c], [b_sub c_sub]);
7 -    I_simplified = simplify(I_exact_result);
8 -    pretty(I_simplified)
```

Output:



Conclusion:

This lab successfully applied core calculus operations—differentiation and integration—to complex exponential and trigonometric expressions. The tasks were solved using the Symbolic Math Toolbox in MATLAB, demonstrating its capability for performing exact analytical calculus (like definite integration and symbolic differentiation) and precisely evaluating the resulting expressions by substituting fixed numerical constants. The work confirmed the bridge between abstract mathematical rules and computational verification.